MTH213 | MAT213 - Elementary DIfferential Equations

Week 9: Techniques for Solving Second-Order Non-Homogeneous Differential Equations

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Introduction

A second-order non-homogeneous differential equation is of the form

$$ay'' + by' + cy = r(x), \qquad r(x) \neq 0.$$

The general solution is:

$$y(x) = y_c + y_p$$

- y_c: complementary solution obtained from the homogeneous equation
- y_p: particular solution depending on the form of $r(x)$

We will learn:

- Method of Undetermined Coefficients
- Method of Variation of Parameters

Learning Outcomes

At the end of the lesson, you should be able to:

- Identify second-order non-homogeneous ODEs.
- Solve for y_p using the Method of Undetermined Coefficients.
- Solve for y_p using Variation of Parameters.

General Solution of Non-Homogeneous ODE

Consider:

$$ay'' + by' + cy = r(x), \qquad r(x) \neq 0.$$

General Solution:

$$y(x) = y_c + y_p$$

- Solve the homogeneous equation to get y_c.
- Find any particular solution y_p to the non-homogeneous equation.

When Can We Use the Method of Undetermined Coefficients?

This method works when the forcing function $r(x)$ is any of the following:

- A polynomial (e.g., $x^2 + 3x$)
- An exponential function (e.g., e^{3x})
- A sine or cosine function (e.g., $\sin 2x$, $\cos 5x$)
- A product of the above types (e.g., xe^{2x} or $e^x \sin x$)
- A **sum** of these functions (e.g., $e^{2x} + 10x$ or $\sin 5x + e^{3x}$)

  (Handle each term separately and add the particular solutions.)

Key Idea:
Guess a suitable form for y_p based on the type of $r(x)$, substitute into the ODE, and solve for the unknown constants.

Problem-Solving Strategy

Steps for Using the Method of Undetermined Coefficients:

1. Solve the homogeneous equation to obtain y_c.
2. Guess an appropriate form for the particular solution y_p based on the type of $r(x)$.
3. Check for duplication with y_c. If any term of y_p appears in y_c, multiply the entire guess by x (or a higher power if necessary).
4. Substitute y_p, y_p', and y_p'' into the differential equation and solve for the unknown constants.
5. Write the general solution:

$$y = y_c + y_p.$$

Case 1: $r(x)$ is a Polynomial

If the forcing function is a polynomial:

$$r(x) = a_n x^n + \cdots + a_1 x + a_0,$$

Then assume a polynomial of the same degree for y_p:

$$y_p = A_n x^n + \cdots + A_1 x + A_0.$$

Example: If $r(x) = 5x^2$, then

$$y_p = Ax^2 + Bx + C.$$

Worked Example 1

Solve the differential equation:

$$y'' + 4y' + 3y = 3x$$

Worked Example 1: Step 1

Step 1: Solve the homogeneous equation.

$$r^2 + 4r + 3 = 0$$

$$(r+1)(r+3) = 0$$

$$r = -1, \quad r = -3$$

Thus,

$$y_c = C_1 e^{-x} + C_2 e^{-3x}.$$

Worked Example 1: Step 2

Step 2: Guess the particular solution.
The forcing function is:

$$r(x) = 3x,$$

a polynomial of degree 1.

Therefore we assume:

$$y_p = Ax + B.$$

Worked Example 1: Step 3

Step 3: Differentiate and substitute.

$$y_p = Ax + B, \qquad y_p' = A, \qquad y_p'' = 0$$

Substitute into the ODE: $y'' + 4y' + 3y = 3x$

$$0 + 4A + 3(Ax + B) = 3x$$

$$3Ax + (4A + 3B) = 3x$$

Worked Example 1: Step 3 (continued)

Match coefficients:

$$3A = 3 \quad \Rightarrow \quad A = 1$$

$$4A + 3B = 0 \quad \Rightarrow \quad 4(1) + 3B = 0$$

$$B = -\frac{4}{3}$$

$$\Rightarrow y_p = x - \frac{4}{3}$$

Worked Example 1: Step 4

Step 4: Write the general solution.

$$y = y_c + y_p$$

$$y = C_1 e^{-x} + C_2 e^{-3x} + x - \frac{4}{3}.$$

Final Answer:

$$\boxed{y = C_1 e^{-x} + C_2 e^{-3x} + x - \frac{4}{3}}$$

In-Class Activity: Case 1 — Polynomial Forcing Function

Solve the following ODEs:

1. $y'' - 4y' + 4y = 2x^2 + 5x - 3$
2. $y'' + 3y' + 2y = 4x^3 - x + 7$

Hint: Guess a polynomial of the same degree as $r(x)$.

Case 2: $r(x)$ is Exponential

If the forcing function is of the form

$$r(x) = Ce^{kx},$$

Then assume a particular solution of the form

$$y_p = Ae^{kx}.$$

Its derivatives are

$$y_p' = kAe^{kx}, \qquad y_p'' = k^2Ae^{x}.$$

Example: If $r(x) = 6e^{4x}$, then

$$y_p = Ae^{4x}.$$

Worked Example 2

Solve the differential equation:

$$y'' - y' - 2y = 3e^{3x}$$

Worked Example 2: Step 1

Step 1: Solve the homogeneous equation.
Characteristic equation:

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

$$r = 2, \quad r = -1$$

Thus

$$y_c = C_1 e^{2x} + C_2 e^{-x}.$$

Worked Example 2: Step 2

Step 2: Guess the particular solution.
The forcing function is

$$r(x) = 3e^{3x},$$

which is an exponential function e^{3x}.

Since e^{3x} is **not** a solution of the homogeneous equation, we guess a particular solution of the same form:

$$y_p = Ae^{3x}.$$

Worked Example 2: Step 3

Step 3: Differentiate and substitute.

$$y_p = Ae^{3x}, \qquad y_p' = 3Ae^{3x}, \qquad y_p'' = 9Ae^{3x}.$$

Substitute into the ODE:

$$9Ae^{3x} - 3Ae^{3x} - 2Ae^{3x} = 3e^{3x}.$$

Combine terms:

$$(9 - 3 - 2)Ae^{3x} = 4Ae^{3x} = 3e^{3x}.$$

Worked Example 2: Step 3 (continued)

Match coefficients and solve:

$$4A = 3 \quad \Rightarrow \quad A = \frac{3}{4}.$$

So

$$y_p = \frac{3}{4}e^{3x}.$$

Worked Example 2: Step 4

Step 4: Write the general solution.

$$y = y_c + y_p$$

$$\boxed{y = C_1 e^{2x} + C_2 e^{-x} + \frac{3}{4} e^{3x}}$$

In-Class Activity: Case 2 — Exponential Forcing Function

Solve the following ODEs:

1. $y'' - y' - 2y = 7e^{3x}$
2. $y'' + 4y = 5e^{-2x}$

Case 3: $r(x)$ is Sine, Cosine, or a Combination

If the forcing function is of the form:

$$r(x) = C\cos(kx) + D\sin(kx),$$

Then assume a particular solution of the form:

$$y_p = A\cos(kx) + B\sin(kx),$$

where k, A and B are constants.

Its derivatives are:

$$y_p' = -Ak\sin(kx) + Bk\cos(kx),$$

$$y_p'' = -Ak^2\cos(kx) - Bk^2\sin(kx).$$

Worked Example 3

Solve the differential equation:

$$y'' + 4y' + 5y = 7\cos(2x)$$

Worked Example 3: Step 1

Step 1: Solve the homogeneous equation.
Characteristic equation:

$$r^2 + 4r + 5 = 0$$

$$r = \frac{-4 \pm \sqrt{16 - 20}}{2} = -2 \pm i$$

Complementary solution:

$$y_c = e^{-2x}\left(C_1 \cos x + C_2 \sin x\right)$$

Worked Example 3: Step 2

Step 2: Guess the particular solution.
The forcing function is

$$r(x) = 7\cos(2x),$$

a cosine function.

Since $\cos(2x)$ is not part of y_c, we try

$$y_p = A\cos(2x) + B\sin(2x)$$

where A and B are constants.

Worked Example 3: Step 3

Step 3: Compute derivatives.

$$y_p' = -2A\sin(2x) + 2B\cos(2x)$$

$$y_p'' = -4A\cos(2x) - 4B\sin(2x)$$

Substitute into the ODE:

$$\begin{aligned}(&-4A\cos 2x - 4B\sin 2x)\\ &+ 4(-2A\sin 2x + 2B\cos 2x)\\ &+ 5(A\cos 2x + B\sin 2x) = 7\cos 2x.\end{aligned}$$

Worked Example 3: Step 3 (continued)

Combine like terms:

$$\cos(2x) : (-4A + 8B + 5A) = A + 8B$$

$$\sin(2x) : (-4B - 8A + 5B) = -8A + B$$

Set equal to the right-hand side $7\cos(2x) + 0\sin(2x)$:

$$\begin{cases} A + 8B = 7 \\ -8A + B = 0 \end{cases}$$

Solve the system:

$$B = 8A \quad \Rightarrow \quad A + 8(8A) = 7 \quad \Rightarrow \quad 65A = 7 \quad \Rightarrow \quad A = \frac{7}{65}$$

$$B = \frac{56}{65} \qquad \Longrightarrow \; y_p = \frac{7}{65}\cos 2x + \frac{56}{65}\sin 2x$$

Worked Example 3: Step 4

Step 4: Write the general solution.

$$y = y_c + y_p$$

$$y = e^{-2x}(C_1 \cos x + C_2 \sin x) + \frac{7}{65}\cos 2x + \frac{56}{65}\sin 2x$$

Final Answer:

$$\boxed{y = e^{-2x}(C_1 \cos x + C_2 \sin x) + \frac{7}{65}\cos 2x + \frac{56}{65}\sin 2x}$$

In-Class Activity: Case 3 — Trigonometric Forcing Function

Solve the following ODEs for the particular solution y_p:

1. $y'' + 4y = 4\sin 3x$
2. $y'' - 2y' + 5y = 3\cos 2x$

Hint: Guess $y_p = A\cos kx + B\sin kx$.

Case 4: Combinations of Functions

If the forcing function $r(x)$ is a combination (sum or product) of the previous cases,

Then take y_p as a combination of the corresponding types:

- $r(x) = x^2 + e^{3x} \quad \Rightarrow \quad y_p = Ax^2 + Bx + C + De^{3x}$
- $r(x) = e^{3x} + \sin x \quad \Rightarrow \quad y_p = Ae^{3x} + B\cos x + C\sin x$
- $r(x) = x\cos 2x \quad \Rightarrow \quad y_p = (Ax + B)\cos 2x + (Cx + D)\sin 2x$
- $r(x) = x^2 e^{5x} \quad \Rightarrow \quad y_p = (Ax^2 + Bx + C)e^{5x}$

Quiz: Determine the Particular Solution y_p for the following:

1. $4e^{-8x}$
2. $(9x-2)e^{3x}$
3. $x+\cos 2x$
4. x^2e^{2x}
5. 100

Quiz (cont'd)

6. $(9x^2 + 3)e^{2x}$
7. $7xe^{2x}\cos 8x$
8. xe^{2ix}
9. $3x^5e^{2x}$
10. $4x + 10\sin x$

Quiz (cont'd)

11. $e^{3x} \sin 3x$
12. $9x^2 + 1$
13. $7x^2 + e^{2x}$
14. $5x^2 \sin x$
15. $x^3 + x$

Worked Example: Case 4

Solve the differential equation:

$$y'' - 3y' + 2y = x^2 + e^{3x}$$

Worked Example: Case 4 — Step 1

Step 1: Solve the homogeneous equation.
Characteristic equation:

$$r^2 - 3r + 2 = 0$$

$$(r-1)(r-2) = 0$$

$$r = 1, \quad r = 2$$

Thus

$$y_c = C_1 e^x + C_2 e^{2x}.$$

Worked Example: Case 4 — Step 2

Step 2: Guess the particular solution.
The forcing function is

$$r(x) = x^2 + e^{3x},$$

which is a combination of a polynomial x^2 and an exponential e^{3x}.

Since neither part is a solution of the homogeneous equation, we guess a particular solution of the same form:

$$y_p = (Ax^2 + Bx + C) + De^{3x}.$$

Worked Example: Case 4 — Step 3

Step 3: Differentiate and substitute.

$$y_p = Ax^2+Bx+C+De^{3x}, \quad y_p' = 2Ax+B+3De^{3x}, \quad y_p'' = 2A+9De^{3x}.$$

Substitute into the ODE:

$$y_p''-3y_p'+2y_p = (2A+9De^{3x})-3(2Ax+B+3De^{3x})+2(Ax^2+Bx+C+De^{3x}$$

$$= 2Ax^2 + (-6A+2B)x + (2A-3B+2C) + 2De^{3x}.$$

Worked Example: Case 4 — Step 4

Step 4: Match coefficients and solve for constants.
Compare with $r(x) = x^2 + e^{3x}$:

$$2A = 1 \quad \Rightarrow \quad A = \frac{1}{2}$$

$$-6A + 2B = 0 \quad \Rightarrow \quad B = \frac{3}{2}$$

$$2A - 3B + 2C = 0 \quad \Rightarrow \quad C = 1$$

$$2D = 1 \quad \Rightarrow \quad D = \frac{1}{2}$$

So

$$y_p = \frac{1}{2}x^2 + \frac{3}{2}x + 1 + \frac{1}{2}e^{3x}.$$

Worked Example: Case 4 — Step 5

Step 5: Write the general solution.

$$y = y_c + y_p$$

$$\boxed{y = C_1 e^x + C_2 e^{2x} + \frac{1}{2}x^2 + \frac{3}{2}x + 1 + \frac{1}{2}e^{3x}}$$

Next: Apply initial conditions if given to solve for C_1 and C_2.

In-Class Activity: Case 4 — Combination of Functions

Solve the following ODEs for the particular solution y_p:

1. $y'' - y' - 6y = x + e^x$
2. $y'' + 2y' + y = x^2 + \cos 2x$

Hint: Take a particular solution as a combination of guesses for each part of $r(x)$.

Worked Example: Duplication Case

Solve the differential equation:

$$y'' - 3y' + 2y = e^{2x}.$$

Step 1: Solve the homogeneous equation

Characteristic equation:

$$r^2 - 3r + 2 = 0$$

$$(r-1)(r-2) = 0$$

$$r = 1, \quad r = 2$$

Thus

$$y_c = C_1 e^x + C_2 e^{2x}.$$

Step 2: Guess the particular solution

The forcing function is

$$r(x) = e^{2x}.$$

Notice: e^{2x} **is already a solution of the homogeneous equation!**

Rule for duplication: Multiply by x until the guess is independent of the homogeneous solution.

$$y_p = Axe^{2x}.$$

Step 3: Differentiate and substitute

$$y_p = Axe^{2x}, \quad y_p' = Ae^{2x} + 2Axe^{2x}, \quad y_p'' = 4Ae^{2x} + 4Axe^{2x}$$

Substitute into the ODE:

$$y_p'' - 3y_p' + 2y_p = (4Ae^{2x} + 4Axe^{2x}) - 3(Ae^{2x} + 2Axe^{2x}) + 2(Axe^{2x})$$

$$= (4A - 3A)e^{2x} + (4A - 6A + 2A)xe^{2x} = Ae^{2x} + 0 = Ae^{2x}$$

Step 4: Solve for the constant

Compare with RHS:

$$Ae^{2x} = e^{2x} \quad \Rightarrow \quad A = 1$$

$$y_p = xe^{2x}$$

Step 5: General solution

$$y = y_c + y_p$$

$$\boxed{y = C_1 e^x + C_2 e^{2x} + x e^{2x}}$$

In-Class Activity: Case 5 — Duplication

Solve the following ODEs:

1. $y'' - 3y' + 2y = e^{2x}$
2. $y'' + 4y = \sin 2x$
3. $y'' + 4y = x \sin 2x$
4. $y'' - 2y' + y = xe^{x}$

Self-Assessment Exercises: Second-Order Linear ODEs

Instructions: Solve the following ODEs and find the general solution. Show all steps.

1. $y'' - 5y' + 6y = x^3 - 2x + 1$
2. $y'' + 2y' + y = 4x^2 + 3x$
3. $y'' - y' - 2y = 5e^x$
4. $y'' + 4y = 2e^{-x}$

Self-Assessment Exercises (continued)

1. $y'' + 9y = 6 \sin 3x$
2. $y'' - 2y' + 5y = 4 \cos 2x$
3. $y'' - y' - 6y = x^2 + e^x$
4. $y'' + 3y' + 2y = x + \cos 2x$

Self-Assessment Exercises (continued)

1. $y'' - 3y' + 2y = e^{2x}$
2. $y'' + 4y = \sin 2x$
3. $y'' + y = x \cos x$
4. $y'' - 2y' + 2y = xe^{x}$

Hint: Use the method of undetermined coefficients and apply duplication rules when necessary.

References and Further Reading

1. Boyce, W. E., DiPrima, R. C. *Elementary Differential Equations and Boundary Value Problems*, 11th Edition, Wiley, 2017.
2. Zill, D. G., Cullen, M. R. *Differential Equations with Boundary-Value Problems*, 10th Edition, Cengage, 2020.
3. Kreyszig, E. *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2011 — Chapters on ODEs.
4. Simmons, G. F. *Differential Equations with Applications and Historical Notes*, 2nd Edition, McGraw-Hill, 1991.

Tip: Practice all types of forcing functions and combinations to master the method of undetermined coefficients.